Christopher You

PhD · Human-Centered Computing

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Summary

PhD in Human-Centered Computing specializing in multi-agent systems, conversational AI architectures, and production-scale applications. Experienced in designing and deploying agentic platforms integrating LLM pipelines, retrieval systems, web frameworks, and tool-calling workflows. Combines rigorous experimental evaluation with scalable full-stack engineering, with a track record of delivering AI systems that support real users in high-stakes domains. Passionate about building AI systems that are not only sophisticated but impactful – grounded in human-centered design, validated with real users, and driven by research.

Experience

Postdoctoral Fellow
Aug 2026 - Current

Johns Hopkins University

- Postdoctoral fellow in the **Center for Digital Health and Artificial Intelligence (CDHAI)**, a research-focused center that coordinates collaborations across academia, industry, government, and policy organizations to address healthcare challenges.

Research Assistant
Aug 2020 - May 2026

University of Florida

- Broad impacts of research: (1) embodied agents to improve care for psychiatric patients (**NIH R34 Grant**), (2) AI clinical trial enrollment platform for cancer patients (**NIH R24/U01 Grant**), (3) AI patient simulations in university curriculum, and (4) conversational AI and agentic systems to support mental well-being.
- Designed Build-a-Being Yourself, a web-based conversational AI research platform enabling domain experts to rapidly author and deploy agent-based interventions at scale across multiple active studies.
- Developed and evaluated a multi-institutional AI Clinical Trial Enrollment Platform (Beta Link), conducting user studies with 1500+ patients and building evaluation pipelines analyzing model outputs, engagement metrics, and behavioral outcomes.
- Investigated human-AI collaboration through 10+ web-deployed user studies, integrating user-state modeling, retrieval-augmented generation, LLM fine-tuning, and VR technologies into experimental research protocols.
- Mentored undergraduate and graduate students in research methods, system development, and academic writing, directly advising 3 undergraduates to peer-reviewed publication.

Teaching Assistant
Jan 2024 - May 2024

University of Florida

- Designed coursework and projects for CEN4722: User Experience Design, including a semester-long capstone in which 25 student teams conducted user interviews with campus organizations and peers to identify pain points and prototype meaningful application solutions.
- Curated weekly readings to reinforce course objectives and develop critical thinking around human-centered design principles.
- Collaborated with faculty to plan lectures and course activities, held regular office hours, and provided feedback to support 100 students across all stages of the design process.

Research Assistant
May 2022 - Aug 2022

Davidson College

- Conducted in-depth research on embodiment and agency in avatars within VR and desktop environments, leading to two publications.
- Built IK-driven avatar systems to achieve high-fidelity user embodiment using consumer VR tracking.

Software Engineering Intern
May 2020 - Aug 2020

Amazon

- Architected knowledge graph-backed query system spanning Amazon Help Services and Alexa, enabling semantic search across large-scale service documentation.
- Designed and implemented full-stack SPARQL query interface, integrating NLP-driven entity resolution with a structured graph backend.
- Built frontend and API layer to expose graph query results to end users at production scale.

Research Assistant
Oct 2018 - May 2020

University of Minnesota

- Created novel redirection techniques for locomotion in VR, leading to two publications.
- Developed programming skills in C# using Unity to explore practical applications of VR systems.

Technical Skills

Agentic Systems	Multi-Agent Orchestration, Agentic Workflows, Knowledge Graphs, MCPs, Tool-Calling, Human-AI Collaboration, Human-in-the-Loop Frameworks
AI/ML	LLMs, PyTorch, HuggingFace Transformers, RAG, Scikit-Learn, Fine-Tuning, Prompt Engineering, Embeddings, Model Evaluation & Benchmarking, LoRA/PEFT, Hyperparameter Optimization
Infrastructure	AWS, Google Cloud, REST APIs, Scalable Backend Architectures, Distributed Systems, Web Frameworks (Node.js, React, Express, JavaScript), MongoDB, SQL/RDS, CI/CD, Git
Research	Experimental Design, Statistical Modeling (Python, R, SPSS), Qualitative/Quantitative Methods, Power Analysis

Education

Ph.D. in Human-Centered Computing — May 2026
University of Florida — GPA: 3.8

- **Advisor**: Dr. Benjamin Lok
- **Dissertation**: "A Change in Perspective: Exploring Interaction Paradigms in AI and Virtual Human Systems that Engage Users and Promote Mental Health"
- **Awards**: Graduate School Preeminence Award ($36k/yr), Generation NEXT Fellowship ($10k/yr), L3Harris Graduate Scholarship ($2,500), Olson-Mitchell STEM Translational Communications Award ($5,000)

M.S. in Computer Science — Dec 2022
University of Florida — GPA: 3.8

B.S. in Computer Science — May 2020
University of Minnesota — GPA: 3.8

- **Advisor**: Dr. Evan Suma Rosenberg
- **Awards**: Undergraduate Research Opportunities Program Award ($2k/yr)

Publications

Journal Proceedings

[5] **You, C.**, Ghosh, R., Vilaro, M., et al. (2025). Alter egos alter engagement: Perspective-taking can improve disclosure quantity and depth to AI chatbots. *Frontiers in Digital Health, 7*, 1655860.

[4] **You, C.**, Venkatakrishnan, R., Venkatakrishnan, R., et al. (2024). A sense of urgency on the sense of agency: Challenges in evaluating agency and embodiment in VR. *IEEE Trans. Visualization & Computer Graphics*.

[3] **You, C.**, Peck, T., Stuart, J., et al. (2024). What my bias meant for my embodiment: Virtual embodiment in desktop-based VR. *Frontiers in Virtual Reality, 5*, 1251564.

[2] **You, C.**, Ghosh, R., Canales, R., et al. (In review). Social perspective-taking and human-AI collaboration can promote change talk and readiness in agent-delivered motivational interviews. *Computers in Human Behavior*.

[1] **You, C.**, Ghosh, R., Kauer, T., et al. (In preparation). Positive-unlabeled learning for fraudulent data detection in online clinical trial recruitment. *J. American Medical Informatics Assoc.*

Conference Proceedings

[7] Tamboli, D. N. A., Canales, R., **You, C.**, et al. (2026). Experiential knowledge in multi-agent virtual health systems: The value of an expert patient agent. *ACM IVA*, 1–10.

[6] **You, C.**, Ghosh, R., Maxim, A., et al. (2026). Build-A-Being Yourself: A web-based system for authoring embodied conversational agents in healthcare research. *Int'l Conf. on Human-Computer Interaction*, 427–446.

[5] Ghosh, R., **You, C.**, Venkatakrishnan, R., & Lok, B. (2025). The impact of health literacy in a virtual agent intervention addressing barriers to cancer clinical trial participation. *ACM Conf. on Human-Agent Interaction*, 76–86.

[4] Ghosh, R., **You, C.**, Venkatakrishnan, R., & Lok, B. (2025). Exploring the influence of multiple virtual agents for addressing barriers to cancer clinical trial participation. *ACM IVA*, 1–10.

[3] **You, C.**, Benda, B., Rosenberg, E. S., et al. (2022). Strafing gain: Redirecting users one diagonal step at a time. *IEEE ISMAR*, 603–611.

[2] **You, C.**, Ghosh, R., Maxim, A., et al. (2022). How does a virtual human earn your trust? Guidelines to improve willingness to self-disclose to intelligent virtual agents. *ACM IVA*, 1–8.

[1] Ghosh, R., **You, C.**, Canales, R., et al. (In submission). Embodying support: Multiple embodied virtual agents on engagement in health information seeking. *ACM IVA*.

Additional Publications and Presentations

12 posters and abstracts at venues including IEEE VR, ACM UIST, and ACM SUI, covering clinical trial recruitment systems, conversational AI platforms, and VR interaction techniques (full list at christopher.you). Mentored 2 undergraduates to first-author peer-reviewed publications in the UF Journal of Undergraduate Research (2025). Presented research across IEEE and ACM conferences, delivered guest lectures at the University of Minnesota and University of Florida on conversational AI and virtual human systems, and interviewed on AI and mental health for The Independent Alligator.